

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	22-07-2026	Duration:	60 minutes

Code of Conduct

*I **K Madan Mohan Reddy (192411206)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____Byreddy__

Instructions to Students:

"Online Loan Eligibility Prediction and Approval Management Platform"

1. Understand the Problem Statement

1.1 Explain the problem in your own words.

The Online Loan Eligibility Prediction and Approval Management Platform is developed to improve the process of loan application, eligibility prediction, and approval in financial institutions. In the traditional loan approval process, customers submit their applications manually and wait for several days or even weeks to receive a decision. This process requires bank employees to verify documents, analyse financial records, and evaluate the applicant's eligibility. Manual verification consumes a significant amount of time and increases the workload of bank staff.

- Human errors may occur during document verification and eligibility assessment, resulting in incorrect loan decisions.
- In some cases, eligible applicants may be rejected, while ineligible applicants may receive approval due to inaccurate evaluation.
- Customers often do not know whether they meet the eligibility criteria before applying for a loan.
- As a result, many applications are rejected after lengthy verification, causing disappointment and wasting valuable time for both customers and banks.

1.2 Identify the objectives of the proposed system.

The primary objective of the Online Loan Eligibility Prediction and Approval Management Platform is to automate the loan eligibility verification and approval process, making it faster, more accurate, and more efficient. The system aims to reduce the time required to process loan applications by replacing manual verification with an intelligent digital platform. It evaluates applicants based on essential factors such as income, employment status, credit history, age, existing financial obligations, and repayment capacity to determine their eligibility for a loan. The proposed system minimizes human errors and ensures fair and consistent decision-making through data-driven analysis.

- It provides applicants with quick and reliable eligibility results, helping them understand their chances of loan approval before submitting a complete application.
- The platform also allows users to apply for loans online, upload the required documents securely, and track the status of their applications in real time.
- For financial institutions, the system simplifies the loan approval workflow, reduces paperwork, lowers operational costs, and decreases the workload of loan officers.

- It securely stores customer information, protects sensitive data through authentication and encryption, and maintains accurate records of all loan applications

1.3 State the scope of the problem.

The scope of the Online Loan Eligibility Prediction and Approval Management Platform is to provide a digital solution for predicting loan eligibility and managing the loan approval process efficiently. The system covers the complete workflow, from user registration and online loan application to document submission, eligibility prediction, application verification, and final approval or rejection. It evaluates applicants using important factors such as age, income, employment status, credit history, loan amount, repayment capacity, and existing financial commitments. The platform enables applicants to submit their information online, upload supporting documents, and track the status of their applications in real time.

- It also allows administrators and loan officers to review applications, verify customer details, manage records, and generate reports.
- The system aims to reduce manual paperwork, minimise processing time, improve decision accuracy, and ensure secure handling of customer data.
- It supports multiple loan categories, such as personal, home, education, and vehicle loans, while providing a user-friendly interface for both customers and bank staff.

2. Identify Key Stakeholders and Challenges

2.1 Identify all the stakeholders involved in the system.

- ☐ **Loan Applicant.**
- ☐ **Bank Officer.**
- ☐ **Administrator.**
- ☐ **Bank Manager.**
- ☐ **Financial Institution.**
- ☐ **Credit Verification Agency.**
- ☐ **Document Verification Team.**

2.2 Explain the roles and responsibilities of each stakeholder.

- **Bank Officer – Verifies customer details, Reviews applications.**
- **Administrator – Manages User accounts, loan records.**
- **Bank Manager – Approves or rejects financial loan application and monitor loan records.**

- **Financial Institution – Provides loan services, maintain customer records.**
- **Loan Applicant – Register, Submit the loan application**

2.3 Discuss the major challenges and issues faced by the stakeholders.

The stakeholders involved in the Online Loan Eligibility Prediction and Approval Management Platform face several challenges and issues during the loan application and approval process. Loan applicants may struggle with understanding eligibility requirements, submitting complete and accurate information, uploading valid documents, and protecting their personal and financial data. They may also face loan rejection due to low credit scores, insufficient income, or incomplete applications.

- Loan officers often deal with a high volume of applications, making it difficult to process each request quickly while ensuring accuracy.
- They must also identify fraudulent documents and make fair decisions based on the available information.
- Banks and financial institutions face challenges in reducing loan default risks, maintaining regulatory compliance, protecting customer data, and balancing fast loan approvals with proper risk assessment.
- The system administrator must ensure continuous system availability, protect the platform from cyberattacks, manage software updates, and prevent unauthorised access to sensitive information.

3. Analyse the Current Approach

3.1 Describe the existing system or current process.

The existing loan approval system in many financial institutions is primarily based on a manual or semi-automated process, which often consumes significant time and effort. Applicants are required to visit the bank or financial institution to collect and submit loan application forms along with the necessary documents, such as proof of identity, income certificates, bank statements, and employment details.

- Loan officers manually verify the submitted information, review the applicant's credit history, and assess financial eligibility based on predefined bank policies.
- This manual verification process is time-consuming and increases the possibility of human errors, inconsistencies, and delays in decision-making.
- Applicants often have to wait several days or even weeks to receive the final approval or rejection of their loan applications.
- In addition, the existing system involves extensive paperwork, making document management difficult and increasing administrative costs.
- Communication between applicants and bank staff is often slow, resulting in poor transparency regarding the application status

3.2 Identify its strengths and limitations.

The existing loan approval system has several strengths as well as limitations. One of its main strengths is that experienced loan officers can manually evaluate each application and consider exceptional cases that may not be identified by automated systems. The manual verification

process helps ensure that submitted documents are carefully checked for authenticity, reducing the risk of approving fraudulent applications. The existing system also follows established banking policies and regulatory guidelines, which helps maintain compliance with financial regulations. However, the system has several limitations.

- It is highly time-consuming because loan officers must manually verify applicant details and supporting documents.
- The process involves a large amount of paperwork, increasing administrative costs and making record management difficult.
- Human errors and subjective judgement may lead to inconsistent loan approval decisions.

3.3 Analyse the problems or gaps that need to be addressed.

The existing loan approval process has several problems and gaps that reduce its overall efficiency and effectiveness. One of the major issues is the heavy dependence on manual verification, which increases processing time and delays loan approval decisions. The large amount of paperwork makes document management difficult and increases operational costs for financial institutions. Human errors and subjective judgement during the evaluation process can lead to inaccurate or inconsistent loan approval decisions. The existing system also lacks real-time loan eligibility prediction, forcing applicants to wait for several days before receiving a response. Limited transparency in the application process makes it difficult for customers to track the status of their loan requests, resulting in poor customer satisfaction.

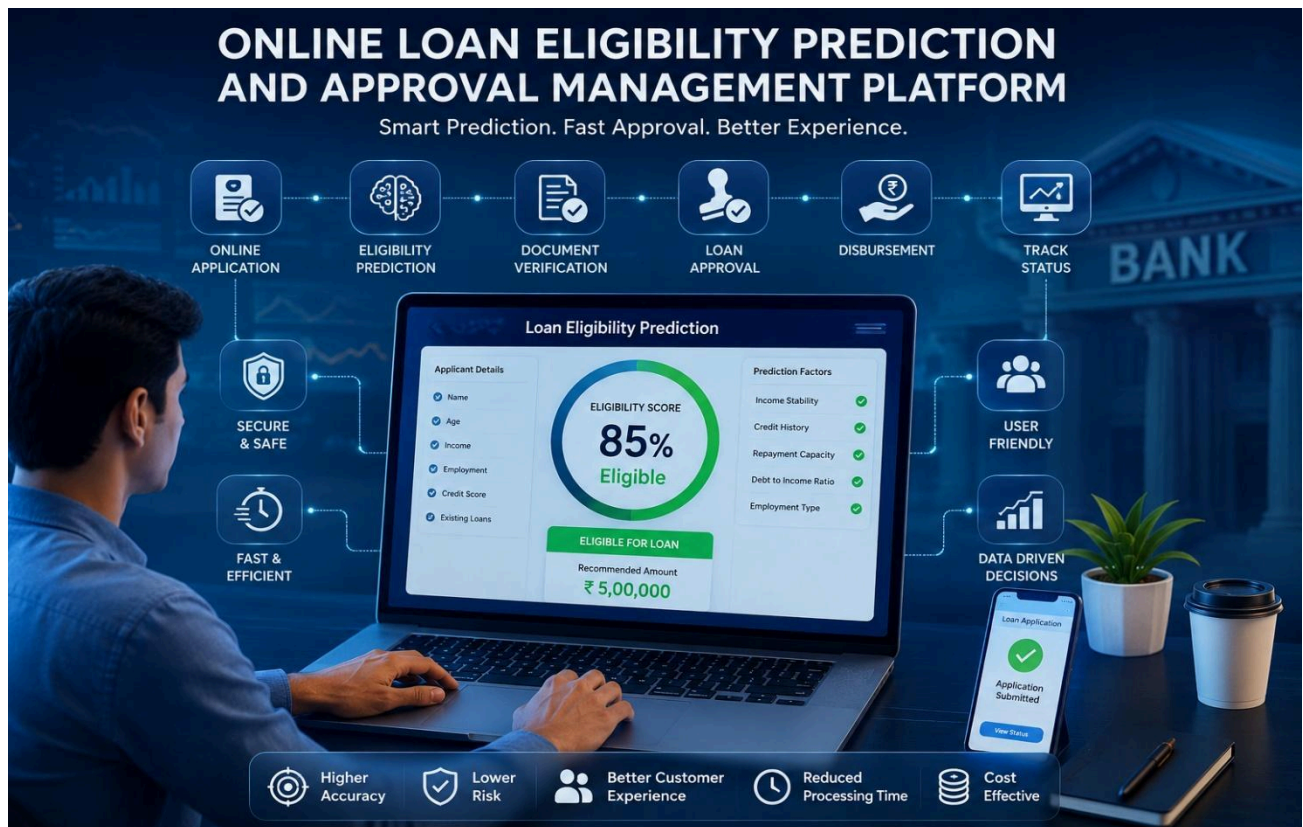
- In addition, the system is not efficient in handling a large number of applications during peak periods, leading to increased workloads for loan officers.
- Detecting fraudulent documents and false information is another significant challenge, as manual verification may not always identify fraud accurately.
- The existing process also has limited data analysis capabilities, making it difficult to assess applicant risk using advanced predictive techniques.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem.

The proposed software-based solution is an Online Loan Eligibility Prediction and Approval Management Platform that automates the entire loan application and approval process. The system allows users to register, log in securely, complete an online loan application form, and upload the required documents through a user-friendly web interface. The platform uses a loan eligibility prediction model to evaluate applicants based on factors such as age, income, employment status, credit score, existing liabilities, loan amount, and repayment capacity. Based on this analysis, the system predicts whether the applicant is eligible for the requested loan. Loan officers can access a dedicated dashboard to review applications, verify documents, and approve or reject loans with the support of the prediction results. Administrators can manage user accounts, monitor system activities, generate reports, and maintain the database securely.

- The platform provides real-time application status tracking, automated notifications, and secure storage of customer information using authentication and data encryption.
- It also maintains a complete history of loan applications for future reference and auditing.



4.2 Describe the major functional modules of the proposed system.

The Online Loan Eligibility Prediction and Approval Management Platform consists of several major functional modules that work together to provide an efficient and secure loan approval process. The User Registration and Login Module allows applicants, loan officers, and administrators to create accounts and securely access the system. The Loan Application Module enables applicants to fill out online application forms, enter personal and financial details, and upload the required documents. The Loan Eligibility Prediction Module analyses applicant information such as income, age, employment status, credit score, existing liabilities, and repayment capacity to predict loan eligibility using predefined rules or machine learning techniques.

- The Document Verification Module allows loan officers to verify the authenticity and completeness of uploaded documents before making a decision.
- The Loan Approval and Management Module enables loan officers to review applications, approve or reject loans, and maintain loan records.
- The Application Status Tracking Module allows applicants to monitor the progress of their loan applications and receive real-time updates.

4.3 Identify the functional and non-functional requirements.

The functional requirements define the core features and operations of the Online Loan Eligibility Prediction and Approval Management Platform. The system should allow users to

register and log in securely using valid credentials. It should enable applicants to fill out online loan application forms, upload the required documents, and submit their applications. The platform should collect applicant information such as personal details, income, employment status, credit score, loan amount, and repayment history for eligibility evaluation. The system should predict loan eligibility based on predefined rules or machine learning algorithms and display the prediction results. Loan officers should be able to review applications, verify documents, approve or reject loan requests, and update application statuses.

- Applicants should be able to track the progress of their applications and receive notifications regarding approval, rejection, or requests for additional documents.
- The administrator should manage users, maintain loan records, generate reports, monitor system activities, and perform database management.
- The system should also securely store customer information, maintain transaction history, and provide search and reporting features for efficient data management.

4.4 Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

Maintainability is ensured through a well-structured architecture, proper documentation, reusable code, and separate modules, making it easier for developers to update, debug, and enhance the system. Reliability is achieved by implementing accurate loan eligibility prediction, proper input validation, error handling, database backup, and recovery mechanisms to ensure continuous and dependable system operation. Usability is incorporated by providing a simple, responsive, and user-friendly interface that allows applicants, loan officers, and administrators to use the platform with minimal training. Features such as online application submission, real-time status tracking, and automated notifications improve the overall user experience.

- Security is one of the most important principles implemented in the system through secure user authentication.
- Role-based access control, password encryption, secure document storage, data encryption, and protection against unauthorised access and cyber threats.

5. Justify the Chosen Methodology/Framework

5.1 Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

For the Online Loan Eligibility Prediction and Approval Management Platform, the Agile Software Development Life Cycle (SDLC) model is the most appropriate choice because it provides flexibility, continuous improvement, and rapid delivery of software features. Agile

divides the project into small development cycles called sprints, where each sprint focuses on designing, developing, testing, and delivering a specific module of the system.

- ❖ This approach allows developers to receive regular feedback from stakeholders such as bank managers, loan officers, and customers, enabling quick modifications based on changing business requirements.
- ❖ Since banking regulations and customer expectations may change over time, Agile supports easy adaptation without affecting the entire project.
- ❖ It also encourages close collaboration among developers, testers, business analysts, and stakeholders, resulting in better communication and higher software quality.

5.2 Justify why the selected methodology/framework is suitable for the proposed system.

The Agile Software Development Life Cycle (SDLC) model is highly suitable for the Online Loan Eligibility Prediction and Approval Management Platform because it supports iterative development, flexibility, and continuous improvement throughout the project. The proposed system consists of multiple independent modules, such as user registration, loan application, eligibility prediction, document verification, loan approval, notification, and report generation, which can be developed and tested incrementally in separate sprints. Agile enables the development team to gather regular feedback from stakeholders, including bank managers, loan officers, administrators, and customers, ensuring that the system meets their requirements effectively.

- ❖ Banking policies, customer needs, and regulatory requirements may change during development, Agile allows modifications to be incorporated quickly without affecting the entire system.
- ❖ Continuous testing and integration help identify and fix defects at an early stage, resulting in higher software quality, improved reliability, and fewer deployment issues.
- ❖ Agile also promotes teamwork, frequent communication, and collaboration among developers, testers, and stakeholders, leading to faster decision-making and better project management.

5.3 Explain how it supports successful project development and maintenance.

The Agile Software Development Life Cycle (SDLC) model supports the successful development and maintenance of the Online Loan Eligibility Prediction and Approval Management Platform by promoting continuous planning, development, testing, and improvement throughout the project. Agile divides the project into short development cycles called sprints, allowing each module, such as user registration, loan application, eligibility prediction, document verification, loan approval, and reporting, to be developed, tested, and delivered incrementally. This approach helps the development team identify and resolve issues early, reducing the risk of major errors during the final deployment.

- ❖ Regular feedback from stakeholders, including customers, loan officers, bank managers, and administrators, ensures that the system meets changing business requirements and user expectations.
- ❖ Agile also encourages close collaboration among developers, testers, and stakeholders, resulting in better communication, faster decision-making, and improved software quality.
- ❖ Continuous testing and integration increase system reliability and minimise software defects.

6. Discuss Expected Outcomes and Limitations

6.1 Describe the expected benefits of the proposed solution.

The system significantly reduces the time required to process loan applications through automated eligibility prediction and digital document verification. It minimises human errors and ensures consistent, data-driven loan approval decisions based on predefined eligibility criteria. The platform eliminates excessive paperwork by allowing applicants to submit forms and upload documents online, thereby reducing administrative costs and improving operational efficiency.

- Applicants can easily track the status of their loan applications in real time and receive instant notifications regarding approvals, rejections, or requests for additional information, leading to greater transparency and customer satisfaction.
- Financial institutions benefit from improved productivity, better resource utilisation, and reduced workload for loan officers, allowing them to focus on complex cases rather than routine verification tasks.
- The system enhances data security through secure authentication, role-based access control, and encrypted storage of sensitive customer information.

6.2 Explain how the solution addresses the identified challenges.

The use of predefined eligibility rules or machine learning algorithms ensures accurate and consistent loan eligibility prediction, reducing human errors and subjective decision-making. Secure user authentication, role-based access control, data encryption, and protected document storage safeguard sensitive customer information and address privacy concerns. The platform also allows applicants to track their application status in real time and receive automatic notifications, improving transparency and customer satisfaction. Loan officers benefit from an organised dashboard that simplifies application review and document verification, reducing workload and increasing productivity.

- The system maintains complete digital records and generates reports that support better monitoring, auditing, and decision-making by bank management.
- Its scalable and modular architecture allows the addition of new features, loan products, and regulatory updates without disrupting existing services.
- Overall, the proposed solution overcomes the limitations of the traditional system by improving speed, accuracy, security, transparency, reliability, and overall efficiency in loan eligibility prediction and approval management.

6.3 Discuss potential risks, implementation challenges, and limitations.

Potential Risks:

1. Risk of data breaches and cyberattacks.
2. Leakage of sensitive customer information.
3. Incorrect loan eligibility predictions due to poor-quality data.
4. Machine learning model bias affecting fair decision-making.
5. System downtime during peak usage.
6. Hardware or server failures.
7. Data loss due to software or database failures.
8. Fraudulent documents uploaded by applicants.
9. Unauthorised access to customer records.
10. Changes in banking regulations affecting system functionality.

Implementation Challenges:

- Collecting accurate and complete customer data.
- Integrating the platform with existing banking systems.
- Ensuring high prediction accuracy.
- Managing large numbers of loan applications efficiently.
- Training bank employees to use the new system.
- Maintaining strong data security and privacy.
- Handling incomplete or incorrect application details.
- Performing regular software updates and maintenance.
- Ensuring compatibility across different devices and web browsers.
- Meeting legal and regulatory compliance requirements.

Limitations:

- Prediction accuracy depends on the quality of input data.
- The system cannot completely eliminate manual verification.
- Machine learning models may require frequent retraining.
- Internet connectivity is required to access the platform.
- Technical failures may temporarily interrupt services.
- Some exceptional loan cases may still require human judgement.
- Fraud detection may not identify every fraudulent application.
- Initial development and deployment costs can be high.
- Regular maintenance is required to ensure system performance.
- The system may need updates whenever banking policies or regulations change.

6.4 Suggest possible future enhancements.

A dedicated mobile application for Android and iOS can increase accessibility and allow users to apply for loans and track their application status from anywhere. The platform can also implement biometric authentication, such as fingerprint or facial recognition, to strengthen security and prevent unauthorised access. Integration with Aadhaar, PAN, and other digital identity verification services can speed up customer verification and reduce fraud. An AI-powered chatbot can provide 24/7 customer support by answering loan-related queries and guiding applicants through the application process. The system can also include advanced fraud detection techniques to identify fake documents and suspicious activities.

- ❖ Cloud-based deployment can improve scalability, performance, and data availability, while blockchain technology can enhance the security and transparency of loan records and document verification.
- ❖ Additional features such as multilingual support, e-signatures, automated notifications, predictive analytics for loan default risk, and interactive management dashboards can further improve the platform.
- ❖ The system can also be expanded to support more loan products, integrate digital payment gateways for loan disbursement and repayment, and continuously adapt to changing banking regulations and customer requirements.

Criteria	Mark
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5